

Dr. Babasaheb Ambedkar Technological University, Lonere, Raigad

A PROJECT REPORT ON

“AI for Legal Case Outcome Prediction”

Project ID: (CSE-AI-11/2025-26)

For the Partial Fulfillment

**Bachelor of Technology in Department of Computer Science Engineering
(Artificial Intelligence)**

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CERTIFICATE

This is to certify that the project report entitled “**AI for Legal Case Outcome Prediction**”, which is being submitted by **Amay Lohar, Nikhil Raghuwanshi, Shital Kalekar and Shraddha Dhanwate** as partial fulfilment for the Degree of Bachelor of Technology (Computer Science Engineering –Artificial Intelligence) of **DBATU, Lonere**

This is bonafide work carried out under my supervision and guidance.

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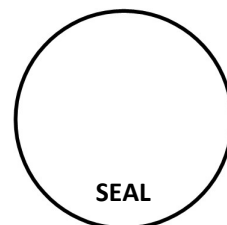
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ACKNOWLEDGEMENT

The satisfaction & euphoria that accompany the successful completion of any task would be incomplete without the mention of people who made it possible. So we acknowledge all those whose guidance and encouragement served as a beacon light & crowned my efforts with success.

We have immense pleasure in expressing thanks to the director Dr. Pramod Patil for providing all the facilities for the successful completion of the project. With due respect, we thank my Head of Department Dr. Prashant Ahire Department of Computer Science Engineering (Artificial Intelligence), for his motivating support, keen interest which kept spirits alive all through.

We would like to express thanks to my guide **Dr. Prashant Ahire, Dr. Priyanka Patil** Department of Computer Science Engineering (Artificial Intelligence) who has guided us throughout the completion of this project.

Finally, we would like to thank all the teaching and non-teaching staff and all our friends who have rendered their support in the completion of this report.

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ABSTRACT

The Indian legal system handles an large volume of cases annually, with judgments often covering hundreds of pages. Consequently, physically analysing precedents and forecasting outcomes is highly complex and time-consuming. This paper presents a production-ready Artificial Intelligence (AI) framework designed to predict Indian Supreme Court case outcomes however addressing modern legal transitions. Our novel architecture utilizes InLegalBERT for deep appropriate surrounding, augmented by a dynamic translation layer that maps legacy Indian Penal Code (IPC) references to the new Bharatiya Nyaya Sanhita (BNS) using structured JSON dictionaries. To overcome the standard 512-token limit of transformers, we fixed a structure-aware chunking pipeline that isolates 'Facts' and 'Arguments' using targeted Regular Expressions. Additionally, to satisfy the essential need for judicial transparency, we integrated SHAP(SHapley Additive exPlanations) with spaCy Named Entity Recognition (NER) to sort out grammatical noise, providing specialists with highly focused, interpretable decision drivers. The entire predictive engine is deployed as a secure SaaS platform utilizing a Streamlit frontend and a Supabase (PostgreSQL) backend for encoded user authentication and case history management. Our results show that this end-to-end system accurately predicts case outcomes and offers brief, explainable justifications, making it highly feasible for real-world application in the Indian legal ecosystem.

Index Terms—Predicting the outcome of legal cases, legal reasoning, the Indian legal system, text classification, explainable artificial intelligence, legal document analysis, court judgment analysis, criminal-judicial data analysis, and legal decision support systems

Keywords: *Legal Case Outcome Prediction, Explainable AI (XAI), Legal Document Analysis, InLegalBERT, IPC to BNS Mapping.*

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